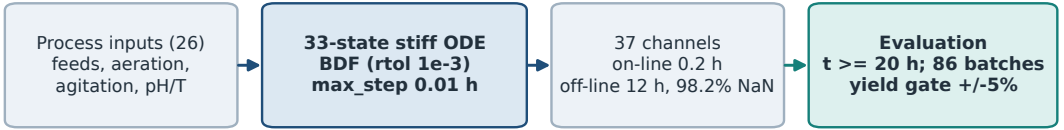
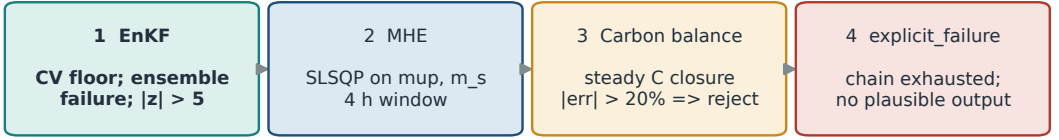
Mechanistic model, observation layers and four-ring estimator fallback



Estimator chain (B/C/D tiers): degradation is explicit, never silent



Observability boundary determined analytically (see Fig. 6):

Residual discovery: sparse regression on d(S_sim - S_v3)/dt gave a death correction beta = (81.70, -293.71, 158.56), R2 = 0.6663, n = 621.

Hierarchical MAP/EM gives only dispersion upper bounds; no independent generalisation error is claimed (the planned hold-out was never reserved).